Linear algebra – new problems

Supplement to the theoretical minimum

Facts needed for the problems in this folder that are not stated in the theoretical minimums of the sheets and cannot be avoided by a solution with the methods of the sheets. Problem numbers refer to *Linear algebra - proposals.pdf*.

Fact 1 (Finite fields). For every prime power q and every $n \geqslant 1$ there is an irreducible polynomial f of degree n over $\mathbb{F}_q$: the number of monic irreducible polynomials of degree n equals $\frac{1}{n}\sum_{d|n}\mu(d)q^{n/d} > 0$. Then $\mathbb{F}_q[x]/(f)$ is a field with q^n elements; via the companion matrix of f it is realised inside $M_n(\mathbb{F}_q)$ as a set of q^n matrices in which every non-zero matrix is invertible. (The sheets state this only for $q = p$ prime.)

Used in: Problem 18.